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VFM-ISRefiner: Towards Better Adapting Vision Foundation Models for Interactive Segmentation of Remote Sensing Images

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***Abstract*—Interactive image segmentation (IIS) plays a critical role in generating precise annotations for remote sensing imagery, where objects often exhibit scale variations, irregular boundaries and complex backgrounds. However, existing IIS methods, primarily designed for natural images, struggle to generalize to remote sensing domains due to limited annotated data and computational overhead. To address these challenges, we propose VFM-ISRefiner, a novel click-based IIS framework tailored for remote sensing images. The framework employs an adaptation tuning strategy that preserves the general representations of Vision Foundation Models while enabling efficient learning of remote-sensing-specific spatial and boundary characteristics. A hybrid attention mechanism integrating convolutional local modeling with Transformer-based global reasoning enhances robustness against scale diversity and scene complexity. Furthermore, an improved probability map modulation scheme effectively incorporates historical user interactions, yielding more stable iterative refinement and higher boundary accuracy. Comprehensive experiments on six remote sensing datasets, including iSAID, ISPRS Potsdam, SandBar, NWPU, LoveDA Urban and WHUBuilding, demonstrate that VFM-ISRefiner consistently outperforms state-of-the-art IIS methods in terms of segmentation accuracy and efficiency. It reduces the average number of clicks (NoC) required to reach 85% and 90% IoU by 0.352 and 0.242, respectively. The proposed method achieves a peak mIoU of 93.87% on NWPU and surpasses 90% mIoU on four datasets with only 8–10 clicks. These results demonstrate the effectiveness and generalizability of our framework, making it highly suitable for high-quality instance segmentation in practical remote sensing scenarios. The codes are available at https://github.com/wondelyan/VFM-ISRefiner .**

***Index Terms*—Remote Sensing Imagery, Interactive Image Segmentation, Vision Foundation Models, Adaptation Module, Probability Map Modulation, Hybrid Attention**

I. Introduction

INTERACTIVE Image Segmentation(IIS) is an effective approach that enables high-precision, pixel-level annotation with minimal human interaction. By continuously incorporating user feedback during the segmentations, such as bounding boxes[1], clicks[2][3] and scribbles[4], IIS can accurately delineate object boundaries and has been widely applied in tasks including medical image analysis, natural scene understanding and remote sensing image interpretation. Compared with fully automatic segmentation methods, IIS offers higher controllability and flexibility, demonstrating significant advantages when handling ambiguous or fine-grained targets[5]. For example, in remote sensing imagery, IIS can effectively distinguish shadow-covered buildings from adjacent vegetation or identify irregularly shaped sandbars within rivers, with only a small amount of user interaction required to correct model prediction errors.

In recent years, the emergence of Vision Foundation Models (VFMs) has ushered in new opportunities for the field of Interactive Image Segmentation (IIS). Vision Foundation Models (VFMs) are typically built upon large-scale pretrained Convolutional Neural Networks (CNNs) and Transformers (e.g., ResNet[6], ConvNeXt[7][8], ViT[9][10][11], and Swin Transformer[12][13]), possessing rich transferable representations learned from massive and diverse datasets[14][15]. Currently, in the field of IIS, numerous studies such as RITM[2], SimpleClick[16] and MFP[17] adopt VFMs as feature extraction encoders, and design additional downstream decoders to complete feature parsing and segmentation mask generation, as showed in Fig.1(a). However, most existing VFMs serving as feature encoders are pretrained exclusively on large-scale natural images, and their performance degrades significantly when directly applied to remote sensing images without further training. For instance, although SAM[18] and its variants demonstrate superior performance on natural images, they require numerous corrective clicks to handle RS scenarios such as small vehicles segmentation(sensitive to scale variation) or shadow-covered buildings(with noisy boundaries), leading to substantially increased interaction costs[19]. To address this issue, most existing IIS methods perform task-specific full-parameter fine-tuning on the encoders of VFMs for interactive segmentation of remote sensing targets. Nevertheless, training VFMs consumes extensive computational resources, and full-parameter fine-tuning may overwrite the general visual knowledge acquired during the pre-training phase, thereby reducing the models' inherent generalization capability[20]. Additionally, annotated data in the remote sensing domain is scarce and costly to obtain[21], which limits the improvement of model's generalization in this specific domain[22].

With the advancement of multimodal large models, several fine-tuning adaptation algorithms (e.g., Adapter[23] and Prompt[24]) have also evolved. By leveraging fine-tuning mechanisms, pretrained backbone networks can be efficiently

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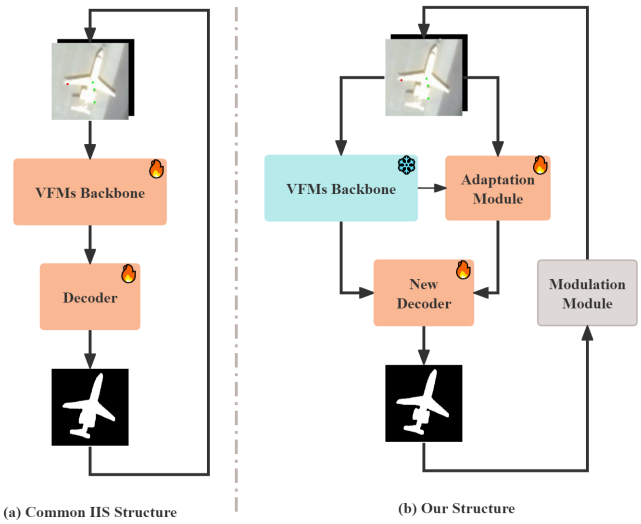


Fig. 1: **Structure Comparison.** (a) represents the common IIS structure(SimpleClick, PiClick, MFP etc.). (b) is the simplified structure used in our paper.

transferred to downstream tasks, reducing training costs while preserving general visual knowledge[25]. For interactive segmentation in remote sensing, leveraging VFMs enables the preservation of general visual priors while selectively optimizing high-level features to better align with the spatial characteristic of RS imagery, as showed in Fig.1(b). For instance, adaptation modules can be employed to capture domain-specific edge structures of RS targets such as the linear boundaries of building rooftops or the elongated contours of roads, without altering the parameters of the pretrained backbone network[26].

However, existing IIS frameworks still face two major limitations when applied to remote sensing imagery. First, most methods(e.g., SimpleClick[16] and AdaptiveClick[27]) fail to fully exploit the historical information of user interactions. Although some approaches(e.g., MFP[17]) attempt to propagate such information through probability map modulation, conventional modulation mechanisms with fixed circular modulation windows are not well-suited to the irregular shapes of remote sensing targets. As a result, prediction instability or oscillation often occurs during multi-round interactions, particularly when dealing with geometrically complex objects such as river sandbars. Second, although VFMs provide powerful visual priors, directly transferring them to remote sensing imagery often leads to degraded boundary accuracy due to domain discrepancies. The visual features learned from natural images (e.g., object textures and edge details) do not align well with the spectral and spatial characteristics of remote sensing objects (e.g., spectral reflectance and spatial configurations)[28]. Moreover, existing adaptation modules such as the Adapter in SAM-HQ[29] are not specifically designed for remote sensing scenarios, making it difficult to capture the intrinsic spectral-spatial correlations of ground objects[20]. Therefore, there is an urgent need for an IIS framework that integrates an efficient VFM-based adaptation mechanism tailored for remote sensing with a robust historical information propagation strategy, enabling high-precision segmentation with minimal user interaction cost.

In response to the above issues, we proposed a novel click-based IIS framework for remote sensing images: VFM-ISRefiner, incorporating the probability map modulation mechanism and the backbone-freezing adaptation fine-tuning algorithm to address the unique challenges of remote sensing data. Drawing on the insights of MFP[17], we leverage probability map modulation to propagate historical interaction information, thereby enhancing the model's capability to retain object shapes and spatial contexts during iterative processes. With respect to the design of the adaptation algorithm, we introduce a module to capture local spatial correlations and edge features specific to remote sensing objects, eliminating the need to retrain the entire backbone network.

In summary, our work advances remote sensing IIS by:

1. Designing an adaptation-oriented feature extraction module for remote sensing, built upon a frozen pretrained VFM backbone to retain general visual representations. By integrating convolutional local modeling with Transformer-based global attention, the module effectively captures spatial details and spectral-spatial correlations, achieving efficient domain adaptation with reduced training cost and improved edge segmentation under complex backgrounds and varying scales.
2. Designing a custom transformer decoder that fuses the global semantic features of vision foundation models with locally enhanced spatial features via bidirectional cross-attention, which effectively integrates global contextual consistency and high-precision boundary details to yield accurate segmentation predictions for remote sensing targets.
3. Adapting an enhanced probability map modulation algorithm to propagate historical interaction information, thereby reinforcing the retention of object shape and spatial context in remote sensing scenarios.

The remainder of this paper is organized as follows. Section II reviews related work on interactive segmentation and vision foundation model adaptation. Section III presents the proposed VFM-ISRefiner framework in detail. Section IV describes the experimental setup and datasets. Section V reports and analyzes the experimental results. Finally, Section VI concludes the paper and discusses future work.

II. Related Work

A. Interactive Image Segmentation

Fine-grained pixel-level segmentation of remote sensing images typically relies on highly time-consuming and expensive manual annotation. Early interactive remote sensing segmentation methods mainly relied on sparse user prompts (such as clicks, scribbles, or bounding boxes). By transforming these prompts into spatial distance maps as auxiliary inputs to Convolutional Neural Networks (CNNs)[30][3][31], the models were guided to dynamically focus on specific geospatial targets. However, these models focus on achieving target segmentation through a single interaction. Due to the complexity of remote sensing target boundaries and spectral

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characteristics, multiple interactive operations are required to improve the segmentation effect. To alleviate this bottleneck, iterative interactive segmentation has garnered extensive attention as an efficient human-in-the-loop paradigm that supports multiple rounds of user interaction[2]. This insight led to widespread adoption of "history information" as input, forming a pipeline for click-based IIS[32][33][34]. With the shift to transformer architectures, subsequent studies[35][36] introduced self-attention mechanisms to enhance the global context awareness of complex structures (e.g., slender roads, irregular buildings) in aerial imagery, significantly reducing the number of interactions required to obtain high-quality masks.

Although Transformer-based IIS frameworks have significantly improved segmentation performance, their adaptation to remote sensing imagery still requires costly full fine tuning and large annotated datasets. This limitation motivates the exploration of Vision Foundation Models (VFMs), which leverage large-scale pretrained Transformers such as ViT[9] and can be efficiently adapted through fine-tuning strategies for interactive segmentation tasks.

B. Vision Foundation Models and Parameter-efficient Fine-Tuning

Vision Foundation Models (VFMs) have emerged as the core driver of modern computer vision, advancing from conventional CNNs[37][38][6][39][40] and Transformers[9][12] to large-scale pretrained frameworks[41][18][42][7]. Within the remote sensing field, the advent of VFMs, particularly the Segment Anything Model (SAM)[18], has greatly propelled the progress of interactive segmentation. As universal feature extractors, these models deliver superior zero-shot generalization performance. State-of-the-art works[36][43][44] have been dedicated to adapting such large models to aerial and satellite imagery, verifying that multimodal prompt interactions via points, boxes or natural language enable efficient target segmentation.

Despite the promising progress, directly deploying VFMs for remote sensing interactive segmentation is plagued by pronounced domain gaps, intensive computational costs and scarce annotated datasets. Moreover, the unique properties of remote sensing imagery, such as densely clustered small objects and low-contrast targets within intricate backgrounds, further impede existing interactive foundation models from balancing interaction efficiency, inference latency and segmentation boundary accuracy.

To mitigate these drawbacks and narrow the visual domain discrepancy, diverse Parameter-Efficient Fine-Tuning (PEFT) strategies have been proposed[45], mainly including Adapter Tuning[23][46], Prompt Tuning[24][28][47] and Low-Rank Adaptation (LoRA)[48]. These approaches reduce training burdens by inserting lightweight learnable modules while freezing most pretrained backbone parameters. Prompt Tuning features ultra-low parameter consumption but suffers from insufficient adaptability in complex visual scenarios; LoRA achieves high tuning efficiency without extra inference overhead[48], yet its low-rank decomposition limits feature representation capability under large remote sensing data distribution shifts. Comparatively, Adapter Tuning[49][29][50] strikes an optimal balance between computational efficiency and feature expressiveness, realizing task-specified adaptation with minimal parameter costs and stable performance. Remote sensing-oriented extensions[45] further validate the effectiveness of parameter-efficient fine-tuning in remote sensing interactive image segmentation (IIS). Motivated by these studies, this work develops a tailored adaptation module to adapt VFMs for remote sensing scenarios, efficiently tackling the task-specific challenges while balancing segmentation precision and computational efficiency.

III. METHOD

In this section, Subsection A will introduce the overall structure of the model as well as the process of feature extraction and transformation; Subsection B will present the basic architecture of the VFM backbone utilized in this study; Subsection C will elaborate on the detailed design of the high-quality feature extractor, explaining how to extract high-quality spatial features from remote sensing images; Subsection D will present the transformer decoder component of the model; and Subsection E will discuss the improvements made to the probability map modulation algorithm.

A. Overall Structure

To effectively adapt large-scale Vision Foundation Models (VFMs) to remote sensing interactive segmentation tasks, our method adopts a modular backbone-adaptation architecture that enables fine-tuning while preserving the pretrained model's generalization capability. As illustrated in Fig.2, the interactive segmentation model proposed in this paper mainly consists of four components: a frozen VFM Backbone, a High-Quality Feature Extractor, a Transformer Decoder and a Probability Map Modulation Module. The framework follows a recursive segmentation-refinement workflow, in which the model iteratively updates its prediction according to user interactions, thereby achieving progressively refined segmentation results with minimal clicks.

Since this paper adopts and improves the idea of probability map modulation from MFP[17], in each interaction round, as shown in Fig.2(a), the input to the network here consists of the image $I \in \mathbb{R}^{3\times H\times W}$, the previous segmentation mask $M_{prev.} \in \mathbb{R}^{1\times H\times W}$, the previous modulated segmentation mask $M_{mod.} \in \mathbb{R}^{1\times H\times W}$ and the click maps $M_{click} \in \mathbb{R}^{2\times H\times W}$ generated from the positive and negative clicks set. In the case of the initial click, since $M_{prev.}$ and $M_{mod.}$ do not exist, it is necessary to set both masks as all-zero matrices. Furthermore, as illustrated in Fig.2(b), $M_{prev.}$, $M_{mod.}$ and M_{click} are concatenated into a unified tensor $M \in \mathbb{R}^{4\times H\times W}$ and jointly processed with the image I through the frozen VFM Backbone to obtain stable and semantically rich representations. The High-Quality Feature Extractor presented in Fig.2(c) then complements these backbone features with fine-grained spatial and boundary information, which are fused through cross-attention operations to generate enhanced local-global feature maps. To further refine the segmentation prediction, $M_{prev.}$, $M_{mod.}$ and I are subjected to Conv. and XceptionConv. operations to extract the corresponding probability

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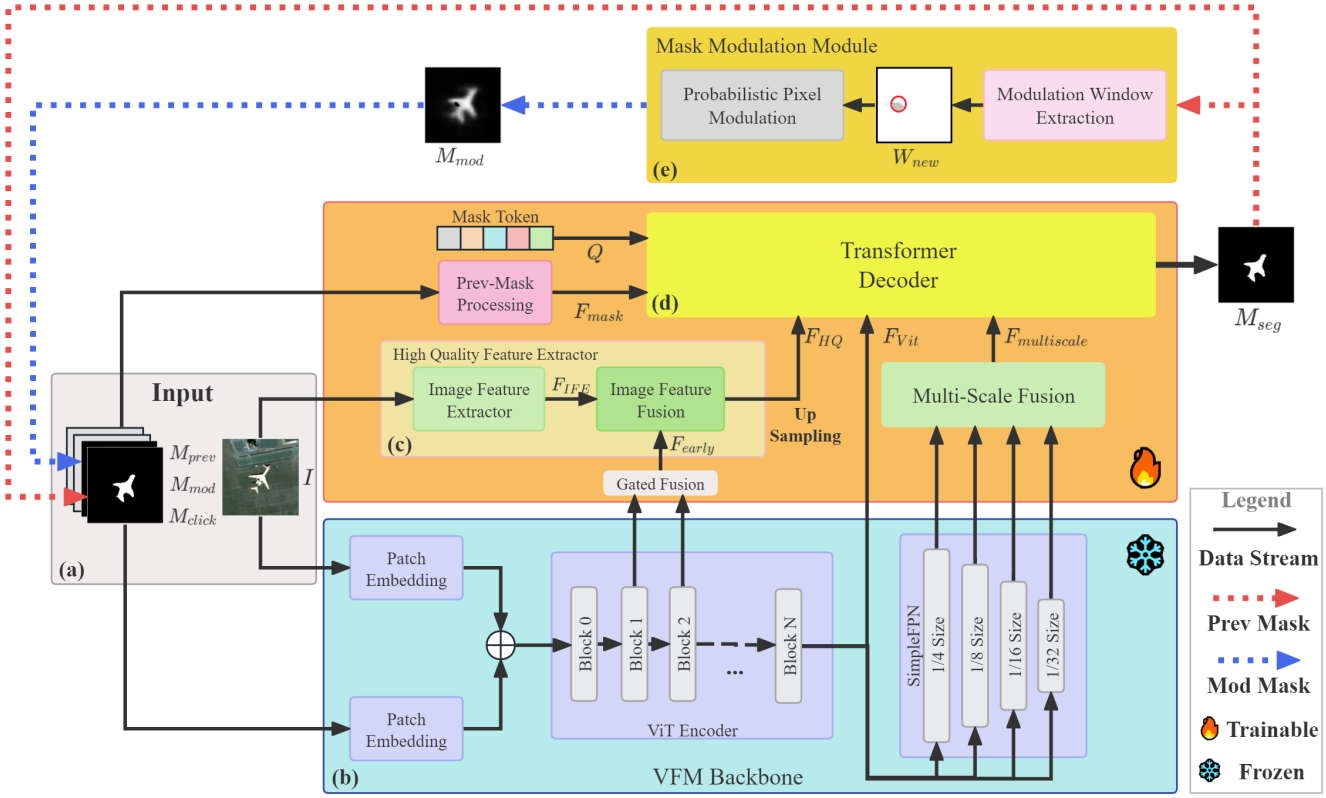


Fig. 2: **Overall Structure.** Our architecture consists of four main components: (b) a frozen Vision Foundation Model (VFM) backbone (blue region), (c) a parameter-trained extractor, (d) a learnable decoder (orange-red region, **learnable modules distinct from MFP**) and (e) a mask modulation module at the top of the diagram (**novel mask modulation design beyond MFP**). Both the training and testing processes follow a recursive workflow.

map feature, which are subsequently fed into the decoder. This paper retains this approach, with the sole modification being that in the input components, image I is substituted with the click map M_{click}, as illustrated in Fig.3. Research findings from [51] have demonstrated that encoding the click map and transmitting it to the decoder contributes to enhancing the model's localization performance. The three inputs are processed by the Prev-Mask Processing Module, ultimately outputting the mask feature F_{mask}. Subsequently, the Transformer Decoder in Fig.2(d) integrates image features, mask features and a learnable mask token to produce an updated instance mask $M_{seg.}$ that balances contextual consistency with edge precision. Finally, as illustrated in Fig. 2(e), the predicted mask is refined through an enhanced probability map modulation mechanism. Specifically, a modulation window is first extracted based on the spatial distribution of user clicks. Subsequently, the pixel probabilities within this localized window are modulated to resolve local inconsistencies, yielding an updated probability map $M_{mod.}$ that is propagated to the next iteration.

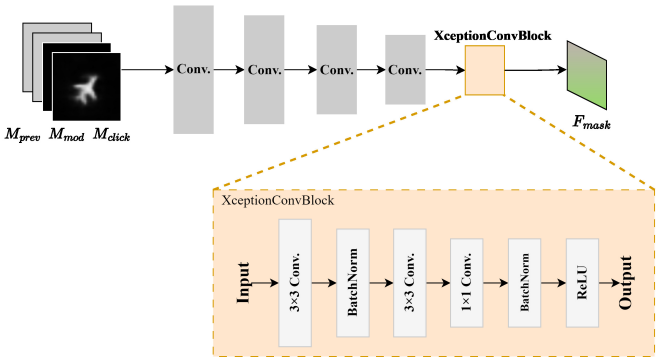


Fig. 3: **Previous Mask Processing Module.** The module is mainly composed of four convolutional modules and one Xception-convolutional module, where the light orange region at the bottom shows the detailed structure of the Xception-convolutional module.

B. Frozen VFM Backbone

To ensure generalization and efficiency, we employ a frozen Vision Foundation Model pretrained on large-scale natural image datasets as the backbone of our framework. Specifically, we utilize the ViT-Based backbone from MFP[17], which provides strong semantic and contextual representations. The backbone parameters remain frozen throughout training, preserving its pretrained knowledge while avoiding overfitting on the limited annotated remote sensing data.

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As illustrated in Fig.2, the VFM backbone consists of three key parts: Patch Embedding, ViT Encoder and SimpleFPN. The image I and concatenated auxiliary tensor M are first processed by the Patch Embedding layer to obtain embedded vectors, which are then passed through the ViT Encoder blocks to extract hierarchical global features. To incorporate local spatial details, we extract the outputs from early ViT blocks F^1_{ViT} and F^2_{ViT}, and fuse them using a learnable gating mechanism:

$$\begin{cases} w_{gate} = \sigma(W \cdot concat\left(F^1_{ViT}, F^2_{ViT}\right) + b) \\ F_{early} = w_{gate} \cdot F^1_{ViT} + (1 - w_{gate}) \cdot F^2_{ViT} \end{cases} \tag{1}$$

where W is the weight matrix in the linear transformation. b is the bias term in the linear transformation. σ denotes the Sigmoid activation function, which serves to map the result of the linear transformation to the interval $[0, 1]$, thereby obtaining the gating weights. The fused feature F_{early} captures mid-level contextual and spatial details, complementing the high-level semantic features extracted by deeper layers. To address the scale variation inherent in remote sensing imagery, the final ViT block output F^N_{ViT} is fed into the SimpleFPN to generate multi-scale representations $F^{ms}_1, F^{ms}_2, F^{ms}_3, F^{ms}_4$, which are fused into a unified multi-scale feature map $F_{multiscale}$. The multi-scale feature will be fed into the trainable transformer decoder, enabling the network to maintain robustness across diverse object sizes.

C. High-Quality Feature Extractor

While the frozen VFM backbone provides rich global semantics and structural priors, these features alone are insufficient to capture fine-grained local variations in remote sensing targets. Therefore, we design a High-Quality Feature Extractor to complement the VFM outputs with detailed spatial representations. This module consists of two components, namely the Image Feature Extractor and the Image Feature Fusion.

As shown in Fig.4, the Image Feature Extractor adopts a residual network architecture, comprising one convolutional residual module, three deformable convolutional residual modules and one linear projection layer. Herein, deformable convolutions are introduced to learn sampling position offsets, enabling adaptive adaptation to variations in the shape, orientation and scale of remote sensing targets. Moreover, the integration of the Squeeze-and-Excitation (SE) Attention mechanism into the deformable convolutional residual modules serves as a widely adopted strategy in convolutional networks to enhance the feature extraction capability of the network. Upon processing image I through the Image Feature Extractor, a spatially informative feature F_{IFE} is derived.

The feature F_{IFE}, together with F_{early} from the early blocks of ViT, is then fed into the Image Feature Fusion module to undergo cross-attention-based fusion. Initially, F_{early} is designated as Query, with F_{IFE} serving as both Key and Value, to execute a F_{early} to F_{IFE} cross-attention operation. The mechanism enables ViT-derived features to integrate partial characteristic information from the convolutional network. Subsequently, the cross-attention output undergoes

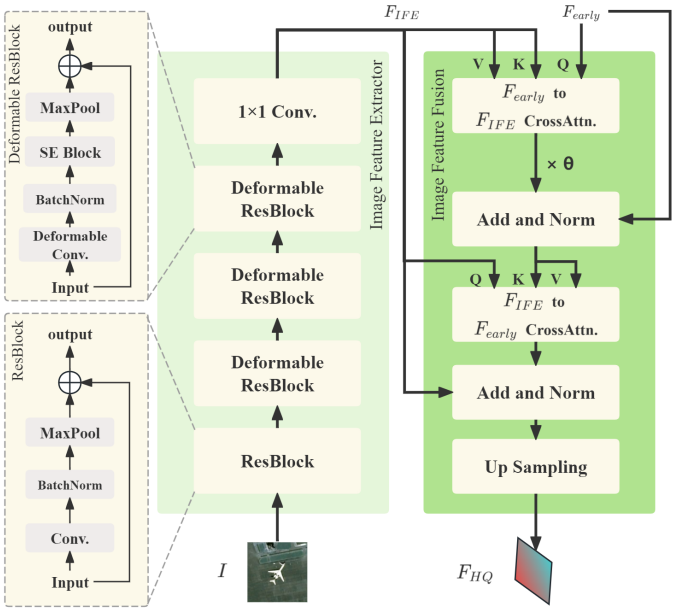


Fig. 4: **High Quality Feature Extractor.** This module comprises an image feature extractor (light-colored region in the middle) and a feature fusion module (green region on the right). The image feature extractor includes one ResBlock, three DeformableResBlocks and one 1×1 convolution, with the detailed architectures of ResBlock and Deformable ResBlock shown on the left of the diagram. The feature fusion module is primarily composed of cross-attention blocks, where θ serves as a learnable parameter.

an element-wise multiplication with a learnable parameter θ which is initialized separately. The resulting product is then added residually to the original F_{early}, followed by a layer normalization operation, as formulated below.

$$F'_{early} = LN(F_{early} + \theta \cdot (CA(F_{early}, F_{IFE}))) \tag{2}$$

where $CA(\cdot)$ is the cross-attention operation, and $LN(\cdot)$ stands for the layer normalization. The reason for introducing the learnable parameter is that the features extracted by ViT contain rich information, and thus this parameter acts to prevent such features from being excessive interference. Subsequently, F_{IFE} is used as Query, while the updated F'_{early} serves as both Key and Value to perform the F_{IFE} to F'_{early} cross-attention operation. This process enables the feature F_{IFE} extracted from convolutional network to assimilate the rich feature information from ViT. The cross-attention output is then added residually to the original F_{IFE}, followed by layer normalization. This sequence of operations preserves information integrity and stabilizes the fusion results, as formalized in the equations below.

$$F'_{IFE} = LN(F_{IFE} + CA(F_{IFE}, F'_{early})) \tag{3}$$

Finally, the fused feature F'_{IFE} undergoes shape adjustment and is upsampled to a high resolution, yielding the high-quality image feature F_{HQ}.

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D. Transformer Decoder

Built upon the high-level semantic features extracted by the frozen Vision Foundation Model, the decoder establishes effective information interaction between the VFM's global representations and the locally enhanced features from the extractor module. Through bidirectional cross-attention processing, the model establishes efficient information interaction between global semantics and local details, thereby generating prediction results with both global consistency and high-precision boundaries.

As illustrated in Fig.5, First, F_{mask} and F_{ViT}^{N} are concatenated, then undergo channel-wise fusion via the Dense Fusion Convolution, which primarily consists of 1×1 convolution operations, to align dimensions with those of token Q. The resulting features F_{DFC}, in conjunction with Q, are fed into a two-layer bidirectional cross-attention module. Diverging from the conventional Transformer Decoder architecture, we herein reverse the sequence of token self-attention and cross-attention operations. As highlighted in [52], tokens are initially randomly initialized and devoid of image feature information, rendering self-attention operations ineffective for acquiring sufficient information at the initial stage. Thus, the token-to-feature cross-attention operation is performed. Q serves as the Query, with Keys and Values derived from F_{DFC}, to align Q with image features. The output then undergoes residual connection and normalization processing. The formula is as follows.

$$Q' = LN(Q + CA(Q, F_{DFC})) \tag{4}$$

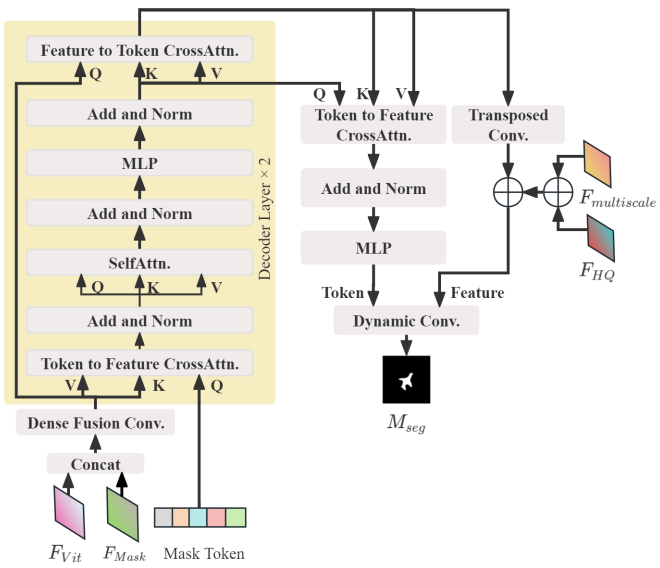


Fig. 5: **Transformer Decoder.** The module consists of a Decoder Layer (the yellow region on the left) and a dual-branch structure (the right region), where the latter includes a Token path and a Feature path. Their outputs are fused via a dynamic convolution module to generate the predicted mask.

Subsequently, Q' undergoes self-attention processing to model long-range dependencies and global context among tokens. This is followed by residual normalization and fully connected processing to maintain training stability. Next, a feature-to-token cross-attention operation is performed, where the previous F_{DFC} serves as the Query, the self-attention processed token acts as both Key and Value. This allows F_{DFC} to receive global semantic supplementation from the tokens, facilitating the understanding of the overall structure by F_{DFC} features. The resulting output F'_{DFC} is directed into two branches: feature branch and token branch. In the feature branch, to incorporate spatial details and address the issue of blurred segmentation boundaries, F'_{DFC} undergoes shape transformation and transposed convolution operations to align their resolution with $F_{multiscale}$ and F_{HQ}. These three features then undergo additive fusion to supplement the detailed information of the image. The above operations are abstracted as shown in the formula.

$$\begin{cases} F'_{DFC} = CA(F_{DFC}, Q') \\ F_{final} = F_{multiscale} + F_{HQ} + TC(F'_{DFC}) \end{cases} \tag{5}$$

where $TC(\cdot)$ represents the transposed convolution operations. In the token branch, a token-to-feature cross-attention operation is reintroduced, where Q' serves as the Query and F'_{DFC} acts as both Key and Value. The output undergoes residual normalization and MLP processing, with the tokens ultimately learning the feature importance weights. The formula is represented as follows.

$$Q_{final} = LN(Q' + CA(Q', F'_{DFC})) \tag{6}$$

Ultimately, a dynamic convolution mechanism is employed to generate the segmentation mask M_{seg}. Specially, convolution kernels and biases are dynamically generated based on the input Q_{final}, and these adaptive parameters are then used to perform convolution operations on F_{final}. In this way, the convolution weights vary with the input content, enhancing the adaptability to different semantics. The relevant formulas are as follows.

$$\begin{cases} [w, b] = MLP(Q_{final}) \\ M_{seg} = w * F_{final} + b \end{cases} \tag{7}$$

where $MLP(\cdot)$ denotes a linear mapping module, where w and b represent the generated convolution kernel weights and biases, respectively, and $*$ denotes 2D convolution.

E. Probability Map Modulation

As noted in [17], probability map modulation enhances or suppresses the probability values of local regions based on the probability map obtained from the previous round and the current user's click information, making them closer to the foreground or background distribution desired by the user. However, in [17], the modulated local region is a circular area, where both foreground and background pixels may coexist. During the probability modification process, a positive click is intended to enhance the foreground probability, but since there is no distinction between foreground and background within this circular region, the background probability is also inadvertently enhanced. To address this issue, we propose an improved probability map modulation algorithm, as illustrated in Fig.6.

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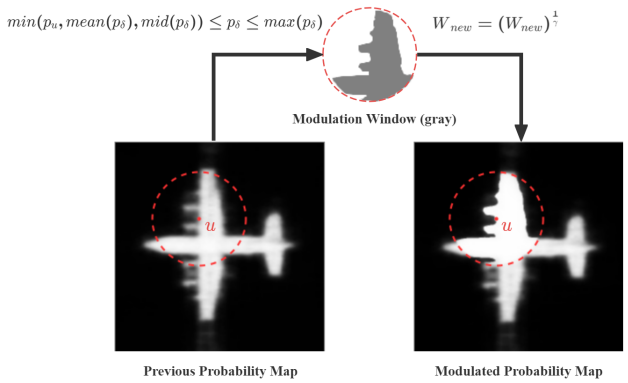


Fig. 6: **Modulate Map Mechanism.** The left image shows the uncorrected result, while the right image presents the corrected version. The modulation window (gray region in the middle image) is extracted using the formula on the left, and the pixel values within this window are refined according to the formula on the right.

After the model outputs the predicted probability map, we obtain the latest click point $u = (x, y, t)$ in the current iteration, where x and y represent the horizontal and vertical coordinates, respectively, and t denotes the click type, with $t = 1$ indicating a foreground click and $t = 0$ indicating a background click. The modulation radius R is then determined based on the click position: if an opposite-type click exists in the vicinity, R is set to half of the distance from the current click to the nearest opposite-type click; otherwise, a predefined radius R_{max} is used. This design prevents the modulation regions of foreground and background from overlapping. To avoid an excessively small modulation radius, we also introduce a minimum radius constraint R_{min}. The equation is given as follows.

$$R = \begin{cases} \frac{1}{2} \cdot min_{c \in \Omega} \|u - c\|, & \Omega \neq \emptyset \\ R_{max}, & \Omega = \emptyset \\ R_{min} & R < R_{min} \end{cases} \tag{8}$$

where Ω denotes the set of clicks of opposite type, $\|\cdot\|$ denotes the Euclidean norm. After obtaining the modulation radius, a circular modulation window is constructed with the click point as the center. Considering that the foreground and background pixel probabilities in the probability map exhibit inherent differences, our method introduces conditional filtering to sieve out irrelevant pixels, extracting only foreground or background regions as the modulation window, which differs from the approach in MFP[17] that modulates the entire circular area. The conditional filtering formula is (9).

In this formula, δ represents a specific pixel point in the image, p_δ $(\delta \in W)$ represents the probability corresponding to pixel point δ, where δ is located within the circular modulation region W, $mean(\cdot)$ represents the average value of the probability pixels in the region, $mid(\cdot)$ refers to the median of the probability pixels in the region, and p_u denotes the pixel probability at the click point. Depending on the click type t_u, pixels whose probability values fall between the minimum and maximum screening thresholds are retained. Subsequently, the γ modulation follows the similar approach of MFP[17]: for different click types, the pixel probabilities in the new modulation region are adjusted, with the adjustment magnitude decreasing as the distance from the click point increases. The relevant formula is (10).

In the formula, $d_\delta = \|\delta - u\|$ represents the distance between the pixel δ and the click u. The modulated probability map will serve as the input for the next iteration of the model. By leveraging the improved probability map modulation algorithm, segmentation errors in target boundary regions can be corrected, thereby enhancing the accuracy and efficiency of interactive segmentation.

IV. Experiments

A. Datasets

To enable the model to effectively accomplish the interactive segmentation task, we selected six high-resolution remote sensing datasets, which were divided into the training set and test set, respectively. Since most of these datasets are originally designed for semantic segmentation tasks, we treated the mask of each closed region in the annotation files as an instance, similar to the datasets used in traditional interactive segmentation tasks.

iSAID[53] is the first benchmark dataset for instance segmentation in aerial imagery. It contains 655,451 object instances across 15 categories in 2,806 high-resolution images collected from multiple sensors and platforms. These images are mostly optical (RGB three-bands) with a few panchromatic ones, having a resolution ranging from 0.5-5m and widths from 800-13,000 pixels. Precise per-pixel annotations for each instance are provided, ensuring accurate localization for detailed scene analysis.

ISPRS Potsdam[54] is a benchmark dataset for remote sensing image semantic segmentation, especially for urban area interpretation. It consists of 38 images with a resolution of 0.05m, all having a width of 6000 pixels and contains near-infrared and RGB bands. The dataset contains six categories: buildings, vehicles as instance categories, and trees, impervious surfaces, low vegetation and background as non-instance categories.

LoveDA Urban[55] is the urban scene segment of the LoveDA dataset. Released by the team from Wuhan University, this dataset contains 1,833 high-resolution optical remote sensing images, all collected in July 2016 from three cities: Nanjing, Changzhou and Wuhan, with the data sourced from Google Earth. The images measure 1024×1024 pixels, provide three RGB channels, and have a ground sampling distance of 30 centimeters. It covers seven land cover categories, including background, buildings and roads. Except for the buildings category, no other categories can be used as instances for experiments. Therefore, in this experiment, buildings are divided into individual instances.

SandBar is a remote sensing instance dataset focus on river sandbars. It is primarily derived from LandSat-8 and GF-2 remote sensing images of the upper reaches of the Yellow River

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$$\begin{cases} W = \{\delta \mid \|\delta - u\| \leq R\}, \\ W_{new} = \{\delta \mid min(p_\delta) \leq p_\delta \leq max(p_u, mean(p_\delta), mid(p_\delta))\}, & t_u = 0 \\ W_{new} = \{\delta \mid min(p_u, mean(p_\delta), mid(p_\delta)) \leq p_\delta \leq max(p_\delta)\}, & t_u = 1 \end{cases} \tag{9}$$

$$\begin{cases} When \quad t_u = 0: \\ \quad \gamma_{max} = \log_{p_u}(0.01), \\ \quad \gamma_\delta = \gamma_{max} \cdot \left(1 - \dfrac{d_\delta}{R}\right) + \dfrac{d_\delta}{R}, \\ \quad \widetilde{W}^\delta_{new} = \left(W^\delta_{new}\right)^{\gamma_\delta} \\ When \quad t_u = 1: \\ \quad \gamma_{max} = \dfrac{1}{\log_{p_u}(0.99)}, \\ \quad \gamma_\delta = \gamma_{max} \cdot \left(1 - \dfrac{d_\delta}{R}\right) + \dfrac{d_\delta}{R}, \\ \quad \widetilde{W}^\delta_{new} = \left(W^\delta_{new}\right)^{\frac{1}{\gamma_\delta}} \end{cases} \tag{10}$$

in China, documenting the conditions of mid-river sandbars in this region. Images in the dataset are three-band optical images with a resolution of 512×512. All annotations and file archiving follow the COCO format. In the experiments, sandbars will be identified as independent instances.

NWPU[56] is an object detection dataset that has been augmented with an instance segmentation version. It is sourced from aerial sensors and primarily consists of optical images (RGB three-band) with a few panchromatic images. The dataset features a resolution range of 0.08-2m across 10 categories, all of which are considered instance categories. In this study, the positive image set is utilized, comprising 650 images with widths varying from 500 to 2000 pixels. The categories are all considered as instance categories in the experiments.

WHUBuilding[57] is a remote sensing image dataset specifically designed for building-related applications. It comprises two main subsets: Satellite Dataset I and Satellite Dataset II. Satellite Dataset I includes 204 high-resolution satellite images captured by sensors such as QuickBird and WorldView, covering various cities globally, with resolutions ranging from 0.3 to 2.5 meters. Satellite Dataset II focuses on East Asia, consisting of 17,388 adjacent satellite images with a uniform resolution of 0.45 meters and a width of 512 pixels, covering 25,749 buildings. In the experiments, one building is considered an instance.

In this experiment, iSAID and ISPRS Potsdam are adopted as the training datasets, encompassing the majority of instance objects in high-resolution images. During the training phase, images from iSAID and ISPRS Potsdam are cropped to a size of 800×800 pixels, with patches lacking annotation masks excluded. Additionally, the ISPRS Potsdam uses RGB bands. The test datasets employed in the experiments consist of iSAID Val, SandBar, LoveDA Urban, NWPU and WHUBuilding, which differ from the training datasets in terms of imaging conditions, resolution and spectral characteristics.

B. Implementation Details

1) Model Details. Since our work is based on the adaptation module set up for one frozen VFM backbone, the parameters of the frozen part adopt the model parameters of MFP[17] trained on the COCOLVIS[2] dataset. The backbone is ViT-Base, with the patch embedding size input to the backbone being (16, 16) and the feature dimension being 768. The output of the backbone is processed by SimpleFPN, and the multi-scale feature dimensions of the output are (128, 256, 512, 1024). The Image Feature Extractor consists of one residual block and three deformable residual blocks, with an input dimension of 3 and an output dimension of 768. In the Image Feature Fusion module, there are two cross-attention layers with 8 heads where both the input and output dimensions are 768. Subsequently, the features undergo an upsampling operation to quadruple the original resolution, resulting in an output dimension of 192. In the Transformer Decoder components, there are two decoder layers with input dimension of 768. Each decoder layer contains two cross-attention modules and one self-attention module, with each attention module having 8 heads. In the dynamic convolution fusion stage of token and feature map, the convolution kernel size is 3×3, with an input dimension of 192 and an output dimension of 1.

2) Training Settings. To generate training data, we combine the iSAID dataset and the ISPRS Potsdam dataset. Images from both datasets are randomly cropped to 448×448. In addition, we perform a series of image augmentation operations, such as random rotation, flipping, brightness adjustment and RGB modification. Regarding the strategies for click points generation and sampling, we adopt the iterative learning framework proposed in RITM[2] and utilize the training sample generation approach introduced in MFP[17] to sample positive and negative clicks. The maximum number of click points per sample is set to 24.

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TABLE I: Comparison of our method with state-of-the-art IIS approaches in NoC80, NoC85 and NoC90 on the iSAID, SandBar, NWPU, LoveDA Urban and WHUBuilding datasets. The best results are highlighted in bold, while the second-best are underlined. * denotes that the backbone parameters of SAM in SAM-HQ are frozen. ** indicates that PiClick was trained without multi-output prediction, using the same model and loss function as the original PiClick.

Method	Backbone	iSAID			SandBar			NWPU			LoveDA Urban			WHUBuilding		
		NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90
RITM[2]	HRNet32	2.37	3.74	6.71	5.11	8.29	13.54	3.25	4.51	7.17	6.73	8.36	10.82	3.82	5.41	9.00
CDNet[58]	ResNet34	5.47	8.14	12.09	8.45	11.76	16.06	5.30	7.84	10.57	9.73	12.58	16.17	7.36	10.20	14.54
FocalClick[32]	HRNet32	6.14	8.28	12.14	8.78	11.85	15.29	5.74	7.78	10.70	9.24	11.84	15.14	7.51	10.06	14.24
GPCIS[59]	ResNet50	3.78	5.84	9.66	8.35	11.34	15.09	3.87	5.67	8.77	7.00	9.18	12.51	5.48	7.65	11.77
SAM-HQ[29]*	SAM-B	10.70	13.47	16.71	11.14	14.98	18.29	11.63	13.31	15.31	15.95	17.71	19.15	11.23	14.40	18.13
OSISeg[45]	SAM-B	10.11	12.13	14.94	12.92	15.44	18.10	10.72	12.59	14.61	14.89	16.44	18.18	11.45	13.42	16.55
SimpleClick[16]	ViT-B	2.40	3.76	6.66	5.25	8.22	13.34	3.08	4.57	7.12	6.82	8.33	10.62	3.64	5.08	**8.42**
PiClick[51]**	ViT-B	3.01	4.61	8.28	5.42	8.64	14.06	4.12	6.06	8.61	8.31	10.16	12.85	3.87	5.35	8.98
AdaptiveClick[27]	ViT-B	2.89	4.61	7.79	6.29	9.37	13.76	3.72	5.36	7.81	6.84	8.44	10.67	**3.54**	**5.03**	8.46
MST[60]	ViT-B	2.77	4.40	8.20	6.15	9.55	14.19	3.08	4.75	7.71	6.84	8.72	11.60	4.16	6.17	10.55
MFP[17]	ViT-B	**2.20**	3.46	6.62	4.63	7.91	12.97	3.24	4.56	7.30	7.13	8.79	11.62	4.01	5.63	9.28
VFM-ISRefiner$_{(ours)}$	ViT-B	2.25	**3.38**	**6.51**	**4.54**	**7.67**	**12.90**	**2.53**	**3.83**	**6.56**	**5.99**	**7.62**	**10.19**	3.65	**5.03**	**8.42**

TABLE II: Comparison of our method against other fine-tuning approaches in NoC85 and NoC90 on the iSAID, SandBar, NWPU, LoveDA Urban and WHUBuilding datasets. The best results are shown in bold. * indicates that the rank of LoRA fine-tuning is 16

	iSAID		SandBar		NWPU		LoveDA Urban		WHUBuilding	
	NoC85	NoC90	NoC85	NoC90	NoC85	NoC90	NoC85	NoC90	NoC85	NoC90
Pretrained Weights	5.40	8.51	8.69	13.24	4.91	7.39	7.66	10.46	6.38	9.97
Full Fine-Tuning	3.46	6.62	7.91	12.97	4.56	7.30	8.79	11.62	5.63	9.28
LoRA[48] *	3.52	6.64	8.14	13.33	4.73	7.41	8.42	11.18	5.23	8.74
Ordinary Adapter[61]	3.57	6.68	8.22	13.44	4.99	7.68	8.58	11.43	5.51	9.07
OSISeg Adapter[45]	3.62	6.84	8.29	13.63	4.84	7.56	8.65	11.28	5.48	9.11
Mona Adapter[46]	3.41	6.55	8.16	13.22	4.82	7.53	8.76	11.34	5.34	8.80
VFM-ISRefiner$_{(ours)}$	**3.38**	**6.51**	**7.67**	**12.90**	**3.83**	**6.56**	**7.62**	**10.19**	**5.03**	**8.42**

During the training process, we apply the Adam optimizer to minimize the normalized focal loss with $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The model is trained for 60 epochs, with the learning rate initially set to 5×10^{-5} and reduced by a factor of 10 at the 10th and 50th epoch. Each epoch comprises 30000 training samples. All models are trained on 2 NVIDIA RTX 4090 GPUs, with the batch size configured to 16. Additionally, we fix the value of the hyperparameter associated with probability map modulation to $R_{max} = 100$, as set in MFP[17].

3) Evaluation Metrics. Adhering to the evaluation strategies proposed by previous works, we adopt average number of clicks (NoC) required to achieve a certain Intersection-over-Union (IoU) ratio. The maximum number of annotated clicks is set to 20. To evaluate performance in high-precision segmentation scenarios, we set the target IoU ratio to 90% and report NoC80, NoC85 and NoC90. Additionally, we plot the mean Intersection-over-Union (mIoU) scores against the number of clicks.

C. Comparative Experiment

In this subsection, we select 11 state-of-the-art interactive segmentation algorithms for comparison with the algorithm proposed in this paper. Among them, RITM[2], CDNet[58], FocalClick[32] and GPCIS[59] adopt convolutional neural networks as their feature extraction backbones. RITM and FocalClick employ HRNet32 as their backbones, while CDNet and GPCIS utilize ResNet34 and ResNet50 as their backbones, respectively; SAM-HQ[29] and OSISeg[45] use a frozen SAM[18] model as its backbone, where the SAM backbone is pretrained on large-scale image datasets. Leveraging the strong feature generalization capability of the backbone, SAM-HQ and OSISeg incorporate an adaptation module to achieve better segmentation performance on specific tasks. SimpleClick[16], PiClick[51], AdaptiveClick[27], MST[60] and MFP[17] all use ViT-B as their feature extraction backbones, with PiClick and AdaptiveClick following a DETR-like architecture.

All the above 11 models are trained on the remote sensing training data mentioned before. The frozen backbone of our model is not fine-tuned on remote sensing images; rather, we directly train only the adaptation module while keeping the backbone parameters fixed at their natural image pretrained values. Correspondingly, the SAM backbone used in SAM-HQ and OSISeg is also not fine-tuned for the remote sensing dataset, and the parameters of SAM backbone are frozen throughout the training process. Apart from this, the other nine models undergo full-parameter training. Additionally, PiClick necessitates hierarchical parent-child relationships among dataset categories to enable the generation of multi-label outputs. However, given the absence of such categorical hierarchies in public remote sensing datasets, we opt to train PiClick using a variant that abstains from producing multiple annotation results.

In addition, this subsection will explore the impact of different fine-tuning methods on model performance, comparing

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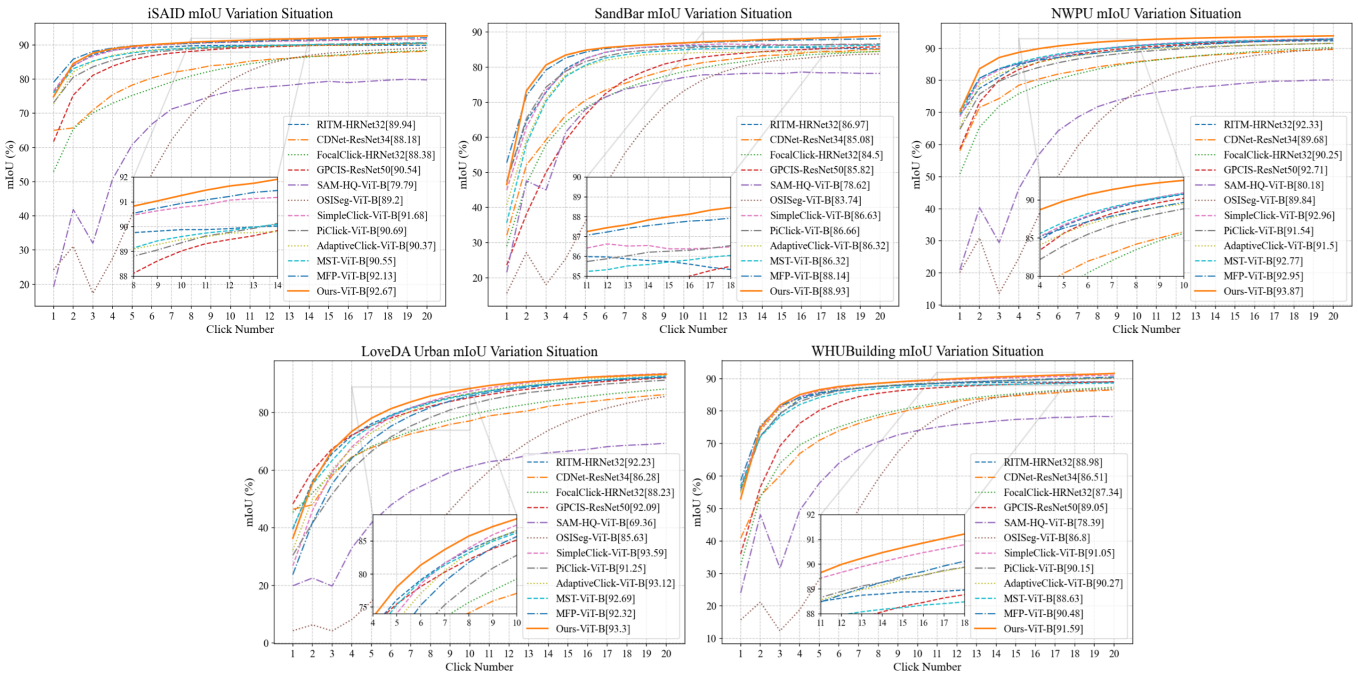


Fig. 7: **mIoU comparison** of our method and other approaches on the iSAID, SandBar, NWPU, LoveDA Urban and WHUBuilding datasets. The horizontal axis denotes the number of clicks, while the vertical axis represents the mean IoU. In the legend, "RITM-HRNET32[89.94]" denotes the RITM model with HRNET32 backbone, which achieves a maximum mIoU value of 89.94% on the dataset.

the segmentation effectiveness and model efficiency among the pretrained model, the fully parameter-trained model, the LoRA fine-tuning model, three adapter fine-tuning methods(Ordinary Adapter, OSISeg Adapter and Mona Adapter) and our proposed model. Notably, the backbone parameters are frozen during the LoRA and adapter fine-tuning process.

1) Comparison of NoC. As shown in Table I, our VFM-ISRefiner method exhibits strong competitiveness and stability overall across multiple remote sensing image segmentation datasets such as iSAID, SandBar, NWPU, LoveDA Urban and WHUBuilding, as well as under different intersection over union (IoU) thresholds including NoC80, NoC85 and NoC90. At the lower NoC80 threshold, our method achieves optimal performance on the SandBar, NWPU and LoveDA Urban datasets, and suboptimal performance on the iSAID dataset. However, at the higher IoU thresholds of NoC85 and NoC90, our method attains the best results on all five datasets. Specifically, the average number of clicks required to reach the IoU 85% and IoU 90% thresholds is reduced by 0.352 and 0.242 respectively compared to that of the suboptimal method. This highlights the distinct advantage of our method in refined interactive segmentation, making it suitable for high-quality segmentation and annotation of instance objects in remote sensing images.

It is noteworthy that SAM-HQ and OSISeg, which adopt a similar technical approach to our method, exhibit poor performance among all models in terms of NoC metric. We attribute this to the fact that SAM-HQ and OSISeg employ backbones pre-trained on natural images, and their pipeline designs are incompatible with recursive interaction patterns. Due to the inherent discrepancies between remote sensing imagery and natural scenes-specifically regarding radiometric characteristics and spectral precision-the generalization capabilities of pretrained backbones are often significantly bottlenecked. Furthermore, the simplistic design of the adaptation modules in SAM-HQ and OSISeg proves insufficient for distilling domain-specific features essential for complex geographical contexts. Beyond these feature-level limitations, both models inherit SAM's decoupled interaction framework, where prompts (clicks or masks) are processed through isolated structures rather than being integrated into the image encoder for early-stage fusion. This architectural separation prevents the model from establishing a deep correlation between image features, iterative clicks, and historical masks. Consequently, the model fails to effectively learn from prior mask errors, rendering it unsuitable for recursive interaction modes and ultimately leading to suboptimal performance in remote sensing scenarios. In contrast, the adaptation module of our method integrates convolutional networks with the global attention mechanism of Transformer, which can compensate for the deficiency of the backbone in extracting features specific to remote sensing images. In addition, the improved probability map modulation method in this paper corrects segmentation errors on target boundaries, and the revised probability map is fused with image features within the encoder, thereby further improving the segmentation accuracy of our model.

2) Comparison of mIoU Variation. As illustrated in Fig.7, we adopt the number of clicks as the independent variable and mIoU as the evaluation metric to conduct a comparative performance analysis of competing methods; specifically, while

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Fig. 8: **Segmentation results** of RITM(HRNet32), GPCIS(ResNet50), SimpleClick(ViT-B), PiClick(ViT-B), AdaptiveClick(ViT-B), MST(ViT-B), MFP(ViT-B) and our VFM-ISRefiner(ViT-B). Red clicks denote negative clicks and green clicks positive clicks. Green regions indicate true positives where predictions overlap with the ground truth contour. Red regions represent false positives and false negatives, i.e., mismatches between the predicted mask and ground truth. The red contour line marks the ground truth mask boundary.

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TABLE III: Efficiency Comparison of Different IIS Methods. † denotes testing time per interaction

Method	Backbone	Trainable Params.	FLOPs	Time †
RITM[2]	HRNet32	30.95M	103.74G	65ms
SAM-HQ[29]	SAM-B	2.09M	55.04G	50.8ms
OSISeg[45]	SAM-B	14.75M	153.31G	96.6ms
SimpleClick[16]	ViT-B	98.03M	169.78G	29.6ms
PiClick[51]	ViT-B	116.28M	197.40G	87ms
AdaptiveClick[27]	ViT-B	121.89M	269.81G	70.6ms
MST[60]	ViT-B	155.52M	239.52G	77.8ms
MFP[17]	ViT-B	98.66M	181.3G	34.2ms
VFM-ISRefiner$_{(ours)}$	ViT-B	40.11M	212.49G	48.2ms

TABLE IV: Efficiency Comparison of Different Fine-Tuning Methods . † denotes testing time per interaction

Method	Trainable Params.	FLOPs	Time †
Pretrained Weights	0M	181.3G	35.4ms
Full Fine-Tuning	98.66M	181.29G	34.2ms
LoRA[48]	1.1M	181.3G	39.8ms
Ordinary Adapter[61]	11.64M	184.99G	53.3ms
OSISeg Adapter[45]	22.33M	197.97G	51ms
Mona Adapter[46]	14.33M	185.46G	58.6ms
VFM-ISRefiner$_{(ours)}$	40.11M	212.49G	48.2ms

the initial mIoU of our proposed method at the first click is not competitive, its mIoU curve exhibits a steep upward trend with increasing click counts, rising sharply within the first 5-10 clicks as reflected in the subfigures corresponding to each dataset, and then rapidly approaches a high-performance plateau that stabilizes after 10-15 clicks. After the final 20th click, our method attains the optimal mIoU among all competing methods on four datasets, namely iSAID (92.67%), SandBar (88.93%), NWPU (93.87%), and WHUBuilding (91.59%), and it still achieves the second-highest mIoU with a value of 93.3% on the LoveDA Urban dataset. Except for the SandBar dataset, the final mIoU values of iSAID, NWPU, LoveDA Urban, and WHUBuilding all surpass 90%, and notably, the iSAID and NWPU datasets, where our method reaches the 90% mIoU threshold within 8 and 10 clicks, respectively demonstrate that our method is capable of rapidly converging to a high mIoU level with a small number of interaction clicks, a characteristic clearly reflected in the curve trends of these two datasets. Collectively, the experimental results across the five datasets indicate that our method possesses robust adaptability and prominent performance advantages in terms of both rapid convergence and high final accuracy, thereby fully validating the effectiveness and precision of the proposed method in click-based interactive segmentation tasks.

3) Comparison With Fine-Tune Methods. As presented in Table II, we compare our method with six baselines: original pretrained MFP, full fine-tuned MFP, LoRA, and three mainstream adapter approaches (Ordinary Adapter, OSISeg Adapter, Mona Adapter). Among them, LoRA, the three adapters and our method all adopt the frozen ViT-B backbone. Across all five remote sensing datasets, our method achieves the best NoC85 and NoC90 performance. It reduces NoC85 from 5.40 to 3.38 on iSAID compared with the pretrained MFP, outperforms full fine-tuning MFP on SandBar and surpasses LoRA Fine-Tuning on NWPU by overcoming its low-rank feature limitation. Compared with the three mainstream adapter methods, our method achieves consistent and significant performance advantages across all datasets: it reduces NoC85 by 0.49-0.62 on SandBar (highly irregular target boundaries) and cuts NoC90 by 0.38-0.69 on WHUBuilding(dense small targets). This performance gap arises from the key limitations of existing adapter designs when applied to remote sensing interactive segmentation scenarios: Ordinary Adapter's simple bottleneck structure cannot effectively capture complex spectral-spatial correlations of remote sensing targets, OSISeg Adapter suffers from structural mismatch when transferred from SAM backbone to ViT-B and insufficient local boundary enhancement, and Mona Adapter optimized for natural image classification lacks targeted design for irregular shapes and complex backgrounds in remote sensing imagery. In contrast, our hybrid convolution-Transformer adaptation module directly addresses these domain-specific challenges, achieving an optimal balance between general knowledge preservation and efficient remote sensing adaptation.

4) Comparison of Efficiency. In this subsubsection of comparisons, we have calculated the number of trainable parameters, FLOPs (Floating Point Operations) of the models, as well as the average time consumed per interaction across the five test datasets. It should be noted that when calculating the average consumed time, all models were tested under identical conditions, including input image size, number of test clicks, hardware specifications for test and so on. As shown in Table III, due to the frozen backbone, the number of trainable parameters in our model is significantly smaller than that of models with ViT-based Vision Foundation Models as backbones which undergo full-parameter training. Consequently, during the model training process, our model involves minimal gradient computations and thus achieves faster training speed. From a comprehensive perspective of FLOPs and average time consumption, although our model exhibits high FLOPs and increased complexity, as it incorporates an additional adapter on top of the Vision Foundation Model, in terms of average time consumption, our model performs favorably, with a shorter average testing time compared to models such as RITM, PiClick and AdaptiveClick. This indicates that our model also holds certain advantages in the generation speed of segmentation results, facilitating efficient segmentation and annotation of remote sensing images.

As shown in Table IV, we calculate trainable parameters, FLOPs and per-interaction inference time for all fine-tuning methods under identical hardware and input conditions. Our method has only 40.11M trainable parameters, which is 59.3% fewer than full fine-tuning (98.66M), leading to much faster training speed. While ensuring sufficient parameter update flexibility to adapt to remote sensing tasks, it greatly reduces the parameter scale compared with full-parameter training, achieving a good balance between parameter efficiency and task adaptability. Compared with the three adapter baselines, our method has slightly higher FLOPs due to the additional High-Quality Feature Extractor and bidirectional cross-attention module. Notably, it achieves the shortest inference

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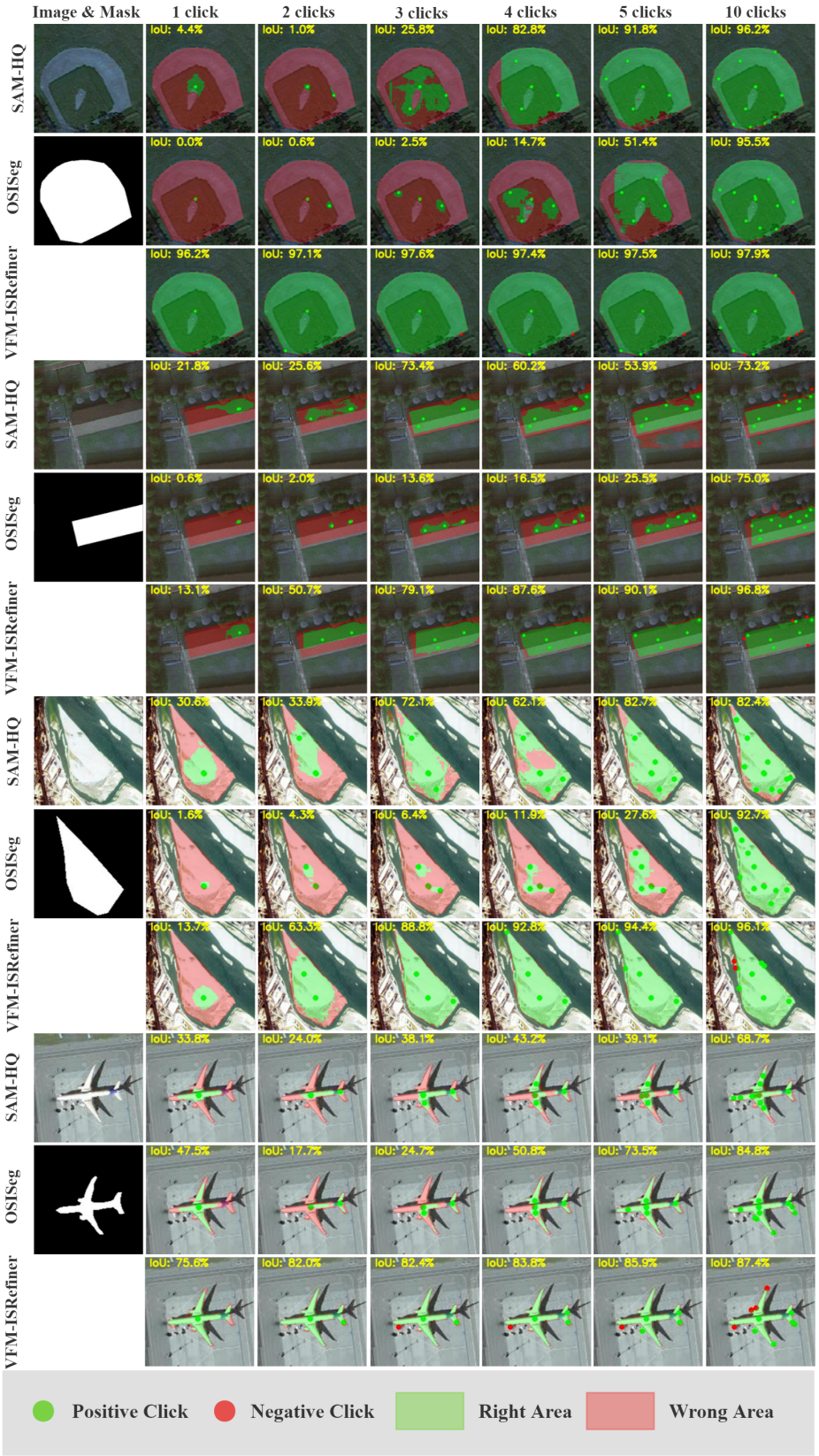


Fig. 9: **Segmentation results** of our method, SAM-HQ and OSISeg. The first column shows the original images and their ground truth masks, while the second to seventh columns illustrate the segmentation results under different numbers of clicks.

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time (48.2ms) among all adapter-based methods, outperforming Ordinary Adapter (53.3ms), OSISeg Adapter (51ms) and Mona Adapter (58.6ms). This result is attributed to our optimized feature pipeline, which extracts compact discriminative features in advance and reduces the computational burden of the subsequent decoder. Although it has slightly higher inference latency than LoRA and full fine-tuning, it delivers far better segmentation accuracy. This trade-off makes it a better choice for practical remote sensing annotation scenarios that prioritize precision with moderate computational overhead.

5) Visualization Comparison. As shown in Fig.8, existing methods often suffer from boundary shifts and regional missegmentation in complex remote sensing scenarios. Traditional convolution-based approaches (e.g., RITM, GPCIS) tend to have over/under-segmentation with complex backgrounds or blurry boundaries, leading to low IoU scores. ViT-based methods (e.g., SimpleClick, PiClick, AdaptiveClick, MST) perform better on samples with regular boundaries (Row 8) but still missegment targets with complex shapes (Rows 1, 2, 6, 9) or under illumination/shadow interference (Rows 7, 10). While MFP optimizes boundary segmentation by refining probability maps, it struggles with small (Row 4) and slender targets (Row 6). In contrast, our improved modulation method achieves stable segmentation across all scenarios, with predictions closely aligning with ground truth and significantly fewer errors. Quantitatively, our IoU exceeds 80% in most cases (even over 90% for some complex targets), whereas other methods generally range from 60% to 80%. This demonstrates our method has stronger robustness and boundary adherence in interactive remote sensing image segmentation-especially for complex boundaries or severe background interference-effectively reducing errors and ensuring accurate, complete results.

Since our method, SAM-HQ and OSISeg are built upon fine-tuning Vision Foundation Models within the similar technical paradigm, we choose to conduct a direct comparison with SAM-HQ and OSISeg, which makes the evaluation more convincing. As shown in Fig.9, SAM-HQ and OSISeg performs poorly with 1-3 clicks, often producing large missegmented regions (Image 1, 3). Though improving with more clicks, it still exhibits noticeable mis-segmentations along complex boundaries and within objects, with final IoU typically 70%-90%. In contrast, our method achieves higher accuracy with fewer interactions. It converges stably as clicks increase, maintaining over 90% IoU across categories. Even for complex targets like airplanes and irregular buildings, our predictions align well with ground truth, with far fewer errors. The above results demonstrate that under a similar technical framework, our method achieves higher segmentation accuracy while requiring fewer user interactions.

D. Ablation Study

To verify the effectiveness and critical of certain components or techniques in the algorithm, we designed four sets of ablation experiments: the role of ViT output features F_{ViT} and high-quality features F_{HQ}, the impact of different values of fusion parameter θ in Fig.4 on the results, the influence of ViT early feature fusion methods on the original mask correction method and the comparison between the original mask modulation method the improved mask modulation method. The experimental conditions remain consistent with those previously used, with the test datasets being iSAID, SandBar and WHUBuilding, and the evaluation metrics being NoC80 and NoC90.

TABLE V: Comparison in NoC80 and NoC90 to investigate the effect of F_{ViT} and F_{HQ} on model performance.

F_{ViT}	F_{HQ}	iSAID		SandBar		WHUBuilding	
		NoC80	NoC90	NoC80	NoC90	NoC80	NoC90
✗	✗	3.03	8.20	6.43	14.16	3.91	9.12
✓	✗	2.79	7.58	6.12	13.95	3.77	8.90
✗	✓	2.29	6.74	4.84	13.24	3.68	8.49
✓	✓	**2.25**	**6.51**	**4.54**	**12.90**	**3.65**	**8.42**

1) The Role of F_{ViT} and F_{HQ}. As can be observed from Table V, the model achieves the poorest performance when both F_{ViT} and F_{HQ} are absent, indicating that these two features exert a certain influence on the model's performance. Specifically, the output of Vision Foundation Model possesses global semantic properties. Feeding this output into the model decoder helps supplement the semantic information of remote sensing images. F_{HQ} generated by the high-quality feature extraction module, incorporates convolutional operations, enabling it to capture local detailed features in images. This facilitates more refined learning of target boundaries and shapes by the model. Additionally, it is evident that the degree of impact on the model differs significantly between the absence of F_{ViT} alone and the absence of F_{HQ} alone. The absence of F_{HQ} substantially degrades the model's performance, whereas the absence of F_{ViT} results in only minor performance loss. This can be attributed to the fact that F_{ViT} output by the ViT, undergoes processing by SimpleFPN to generate $F_{multiscale}$, with $F_{multiscale}$ retaining most of the semantic information from F_{ViT}. Consequently, the absence of F_{ViT} has a limited impact on model performance. Conversely, F_{HQ} represents an additionally extracted high-quality image feature that holds high value for model's feature learning. The lack of this feature leads to suboptimal segmentation results for remote sensing targets with complex boundaries.

TABLE VI: Comparison in NoC80 and NoC90 to analyze the impact of θ on the model performance.

	iSAID		SandBar		WHUBuilding	
	NoC80	NoC90	NoC80	NoC90	NoC80	NoC90
$\theta = 1$	2.28	6.87	5.42	13.50	3.70	8.70
$\theta = 0.5$	2.36	6.79	4.90	13.42	3.97	9.00
Learnable θ	**2.25**	**6.51**	**4.54**	**12.90**	**3.65**	**8.42**

2) The Impact of θ. Table VI presents the results of ablation experiments on parameter θ. When θ is set to a fixed value, the overall performance at $\theta = 0.5$ outperforms that at $\theta = 1$, indicating that θ can balance the performance contributions of F_{IFE} and F_{early} during cross-attention fusion. Although a fixed θ is effective for some situations, it struggles to maintain stable performance across different scenarios. In contrast,

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TABLE VII: Comparison in NoC80 and NoC90 to analyze the impact of F_{ViT}^{early} fusion on the model performance.

	iSAID		SandBar		WHUBuilding	
	NoC80	NoC90	NoC80	NoC90	NoC80	NoC90
only F_{ViT}^{1}	2.36	7.19	5.17	13.54	3.95	8.83
only F_{ViT}^{2}	2.32	6.63	4.93	13.21	3.85	9.08
Fixed Fusion	2.29	6.97	5.06	13.48	3.74	8.56
Gated Fusion	**2.25**	**6.51**	**4.54**	**12.90**	**3.65**	**8.42**

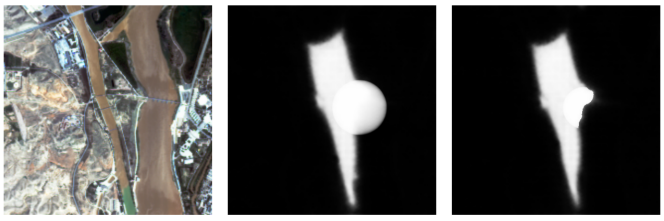


Fig. 10: **Visualization of probability map modulation** using the MFP's algorithm and our proposed algorithm. The left shows the image input, the middle image presents the modulation result of the MFP's algorithm, and the right image illustrates the modulation result of our method.

setting θ as a learnable parameter yields better results across all datasets and settings. This demonstrates that the learnable θ enables the model to adaptively allocate the importance of different feature sources, thereby enhancing overall robustness and generalization capability.

3) The Influence of F_{ViT}^{early} Fusion. Table VII reports the ablation study on different early ViT feature fusion strategies. Using only a single layer feature F_{ViT}^{1} or F_{ViT}^{2} leads to suboptimal performance, indicating that features from a single level are insufficient to capture the multi-scale information required in complex scenarios. Fixed Fusion brings moderate improvements, but its inability to adaptively weight different feature sources results in limited gains. In contrast, our proposed Gated Fusion consistently achieves the optimal performance across all datasets and settings. These demonstrate that Gated Fusion can effectively learn the relative importance of features from different layers, exploit their complementarity, and thereby improve the adaptability and reliability of the model in handling diverse and challenging remote sensing conditions.

TABLE VIII: Comparison of the impact of the MFP's modulation algorithm and our proposed algorithm on the model under the NoC80 and NoC90 metrics.

	iSAID		SandBar		WHUBuilding	
	NoC80	NoC90	NoC80	NoC90	NoC80	NoC90
MFP	**2.24**	6.70	4.79	13.21	3.94	9.11
Ours	2.25	**6.51**	**4.54**	**12.90**	**3.65**	**8.42**

4) The Contrast of MFP's and Our Modulation Method. To validate the effectiveness of the proposed probability map modulation algorithm, we conduct this ablation study. Quantitatively, as shown in Table VIII, on the iSAID, SandBar and WHUBuilding datasets with NoC80 and NoC90 as metrics, our method consistently reduces the error compared to the baseline. For instance, on the SandBar dataset, the NoC80 metric decreases from 4.79 to 4.54, and the NoC90 metric decreases from 13.21 to 12.90. From a qualitative perspective, Fig.10 illustrates that the baseline method produces spurious "circular" response regions in the modulated probability maps, whereas our approach generates results that more closely adhere to the true object contours. This demonstrates the ability of the proposed algorithm to effectively suppress wrong regions in probability maps. Overall, the improved modulation algorithm provides higher-quality probability maps to support subsequent interactive steps, thereby facilitating both enhanced segmentation accuracy and improved interaction efficiency.

E. Limitations and Future Directions

Despite the promising performance achieved by VFM-ISRefiner, several limitations remain. First, the multi-component inference architecture results in moderately higher computational overhead than lightweight CNN-based models, posing minor constraints for deployment on extremely resource-constrained edge devices. Secondly, the current framework is mainly evaluated on optical remote sensing images. Although the adopted datasets cover diverse scenes, spatial resolutions, object categories and geographical conditions, remote sensing data in real-world applications often include additional modalities, such as multispectral, hyperspectral, SAR and LiDAR data. These modalities contain complementary spectral, structural and scattering information, which may be useful for segmenting objects with weak visual contrast or complex material properties.

In future work, we plan to investigate more lightweight and efficient fine-tuning strategies to reduce computational complexity while maintaining segmentation accuracy. We also aim to explore modality-aware adaptation modules and cross-modal feature fusion strategies to better exploit heterogeneous remote sensing data. In addition We will integrate the method into large-scale annotation platforms for practical applications.

V. CONCLUSION

In this work, we propose VFM-ISRefiner, a novel interactive segmentation framework specifically designed for remote sensing images. By incorporating an adaptation module alongside frozen Vision Foundation Model, our method effectively balances general feature representation with domain-specific adaptation, significantly improving boundary precision. Furthermore, the integration of a convolution-transformer hybrid attention mechanism enhances the model's capability to handle large-scale variations and cluttered backgrounds, while the improved probability map modulation algorithm ensures stable propagation of historical interactions for refined segmentation. Extensive experiments on multiple RS datasets, including iSAID, SandBar, NWPU, LoveDA Urban and WHUBuilding, demonstrate that VFM-ISRefiner achieves

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competitive or superior performance compared with state-of-the-art methods in terms of segmentation accuracy, interaction efficiency, and robustness. In particular, our method achieves superior performance under high-precision settings (NoC85 and NoC90), validating its effectiveness in practical annotation scenarios.

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